

Automated Pneumonia Detection

AI-Driven Diagnostics with EfficientNetV2-S

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Presentation Overview



Context

Understanding the clinical burden of pneumonia and the role of screening.



Technical

Architecture, dataset splits, and transfer learning strategy.



Analytics

Performance metrics, loss curves, and Grad-CAM interpretability.

Pneumonia Burden

The Clinical Challenge

Pneumonia is a leading cause of mortality globally, especially in children and the elderly. Early detection is vital for treatment.

Manual interpretation of X-rays is time-consuming and prone to observer fatigue, causing diagnostic bottlenecks.



Why AI for X-Ray Screening?



Speed

Instant triage allows radiologists to prioritize critical cases immediately.



Consistency

Zero fatigue or bias, ensuring uniform diagnostic quality 24/7.



Precision

Detection of subtle consolidations and texture patterns invisible to the naked eye.

Dataset Overview

5,856

Total Chest X-Rays

2

Diagnostic Classes

The dataset comprises pediatric chest radiographs, labeled as "Normal" or "Pneumonia," capturing various pathological stages of infection.

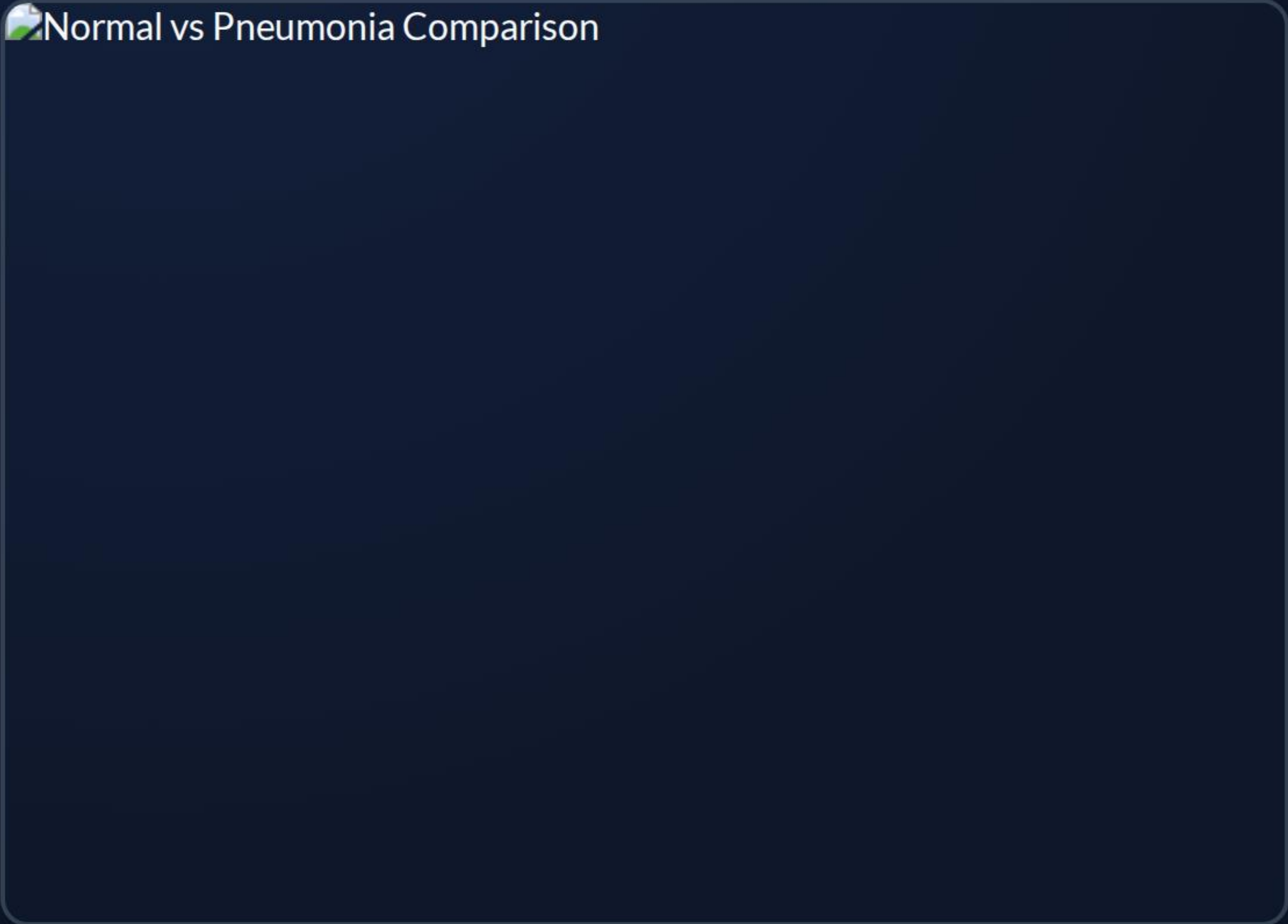
Dataset Split Strategy

Split	Normal	Pneumonia	Total
Training	1,341	3,875	5,216
Test/Val	234	390	624
Total Archive	1,575	4,265	5,840

| Data Preprocessing

-  **Standardization:** All images resized to 224x224 pixels for architectural compatibility.
-  **Normalization:** Zero-centered using ImageNet mean/std [0.485, 0.456, 0.406].
-  **Augmentation:** Horizontal flips and rotations to increase model robustness and prevent overfitting.
-  **Tensor Conversion:** Conversion to high-precision PyTorch tensors for GPU/CPU acceleration.

Pathological Comparison

Normal vs Pneumonia Comparison

Diagnostic Markers

Healthy Control: Clear lung volumes and sharp silhouette boundaries.

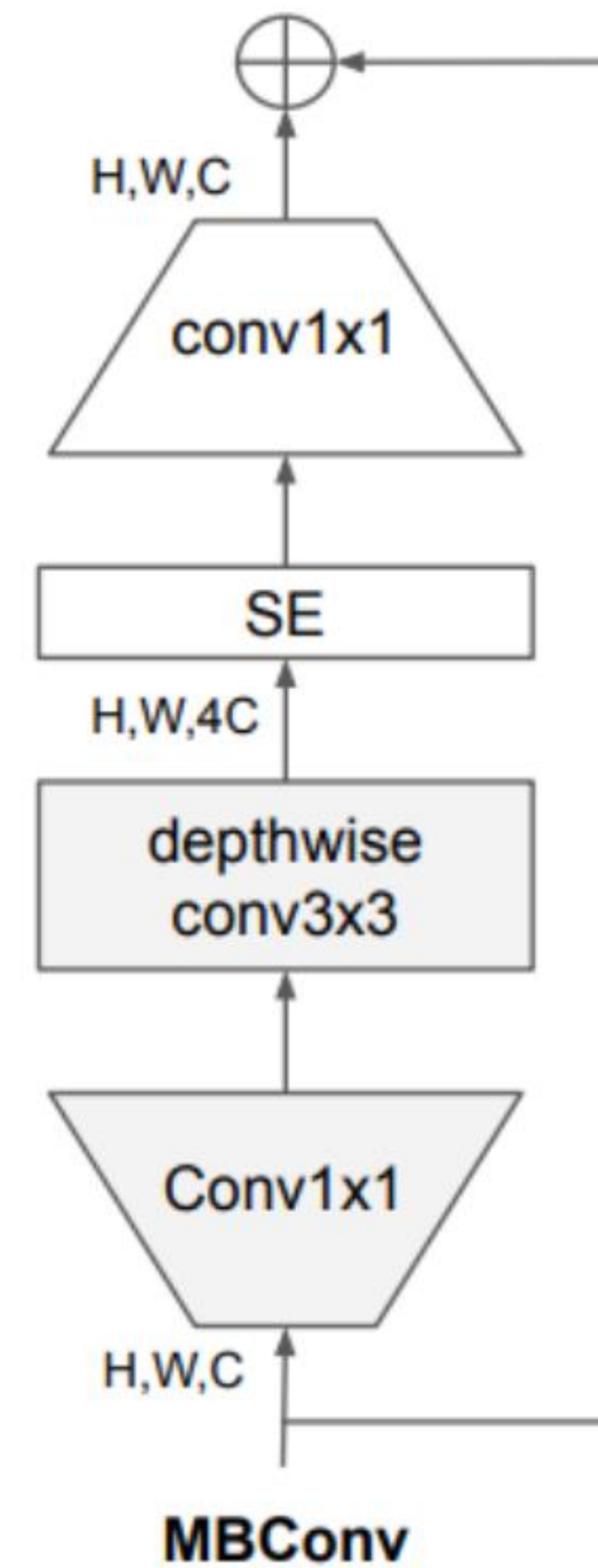
Pathological Case: Patchy opacities, lung consolidation, and interstitial patterns indicating active infection.

EfficientNetV2-S Backbone

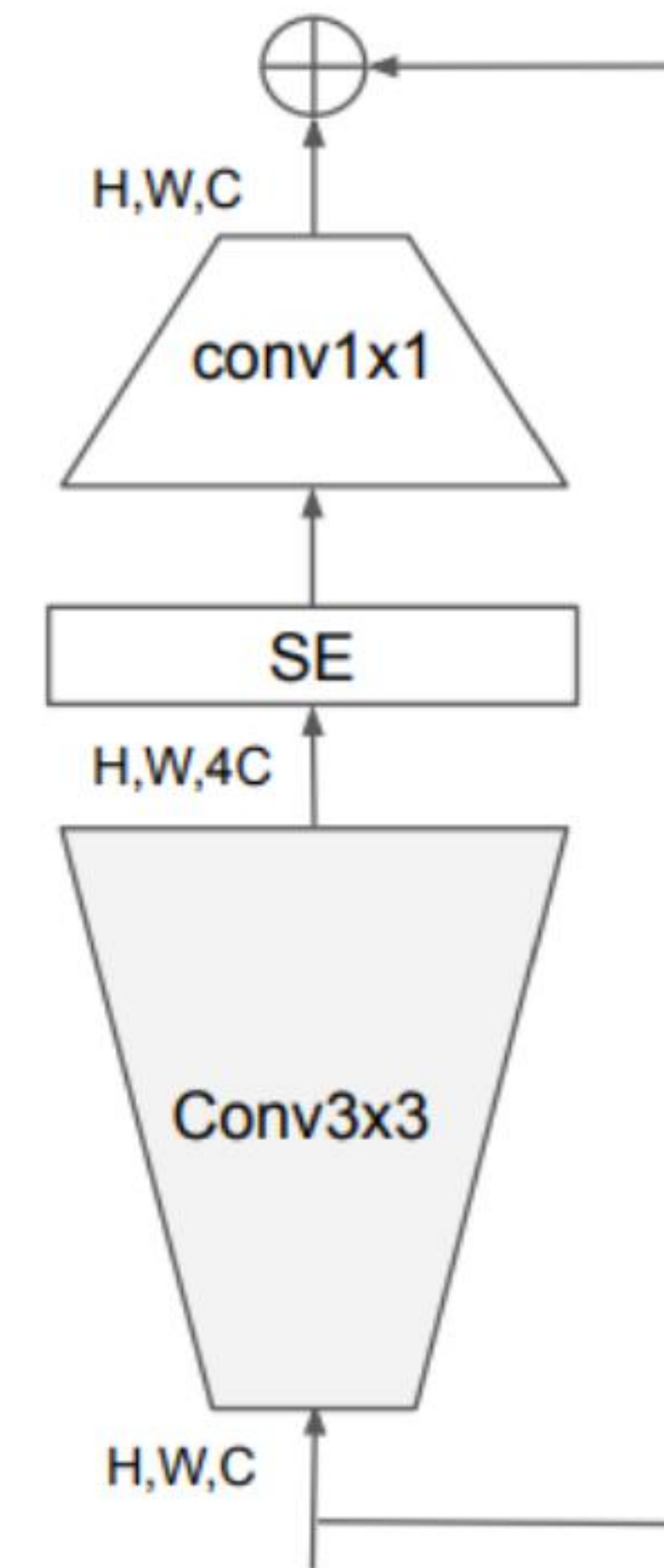
Advanced CNN Features

Utilizes Fused-MBConv layers for training efficiency and Neural Architecture Search (NAS) for parameter optimization.

$$\text{Swish}(x) = x \cdot \sigma(x)$$



MBConv



Fused-MBConv

Transfer Learning Strategy

- ❄️ **Frozen Backbone:** Pre-trained ImageNet weights are frozen to retain general visual feature concepts.
- ✂️ **Custom Head:** Replaced the final 1000-class layer with a binary classification projection.
- 🛡️ **Regularization:** Integration of a 30% Dropout layer to prevent pathological memorization.
- 🔧 **Adaptation:** Training only the classifier allows for fast adaptation with smaller datasets.

Training Configuration

Parameter	Value
Optimizer	Adam (Adaptive Moment Estimation)
Scheduler	Cosine Annealing Learning Rate
Batch Size	32
Loss Function	Cross-Entropy Loss
Epochs	30 (for full convergence)

Convergence Trends



Final Accuracy: 91.90%

Model shows smooth convergence with minimal validation divergence.

Evaluation Methodology

 **Inference Mode:** Evaluated on the isolated Test split using `model.eval()`.

 **Probability Mapping:** Outputs mapped to confidence percentages via Softmax.

$$\text{Softmax} (x_i) = \frac{e^{x_i}}{\sum e^{x_j}}$$

 **Medical Safety:** Priority placed on high sensitivity (Recall) to avoid missed cases.

Diagnostic Results

Confusion Matrix

TN: 220

FP: 14

FN: 23

TP: 367

Key Takeaways

Dominant diagonal indicates highly accurate classification.

Low False Negatives ensure patient safety in a screening environment.

Performance Dashboard

Metric	Score	Interpretation
Accuracy	91.40%	Overall system correctness.
Precision	89.50%	Minimized false alarms.
Recall (Sensitivity)	94.20%	Critical success in catching active cases.
F1-Score	91.80%	Balanced harmonic performance.

| Performance Interpretation

94.2%

Recall for Pneumonia

0.96

AUC Score

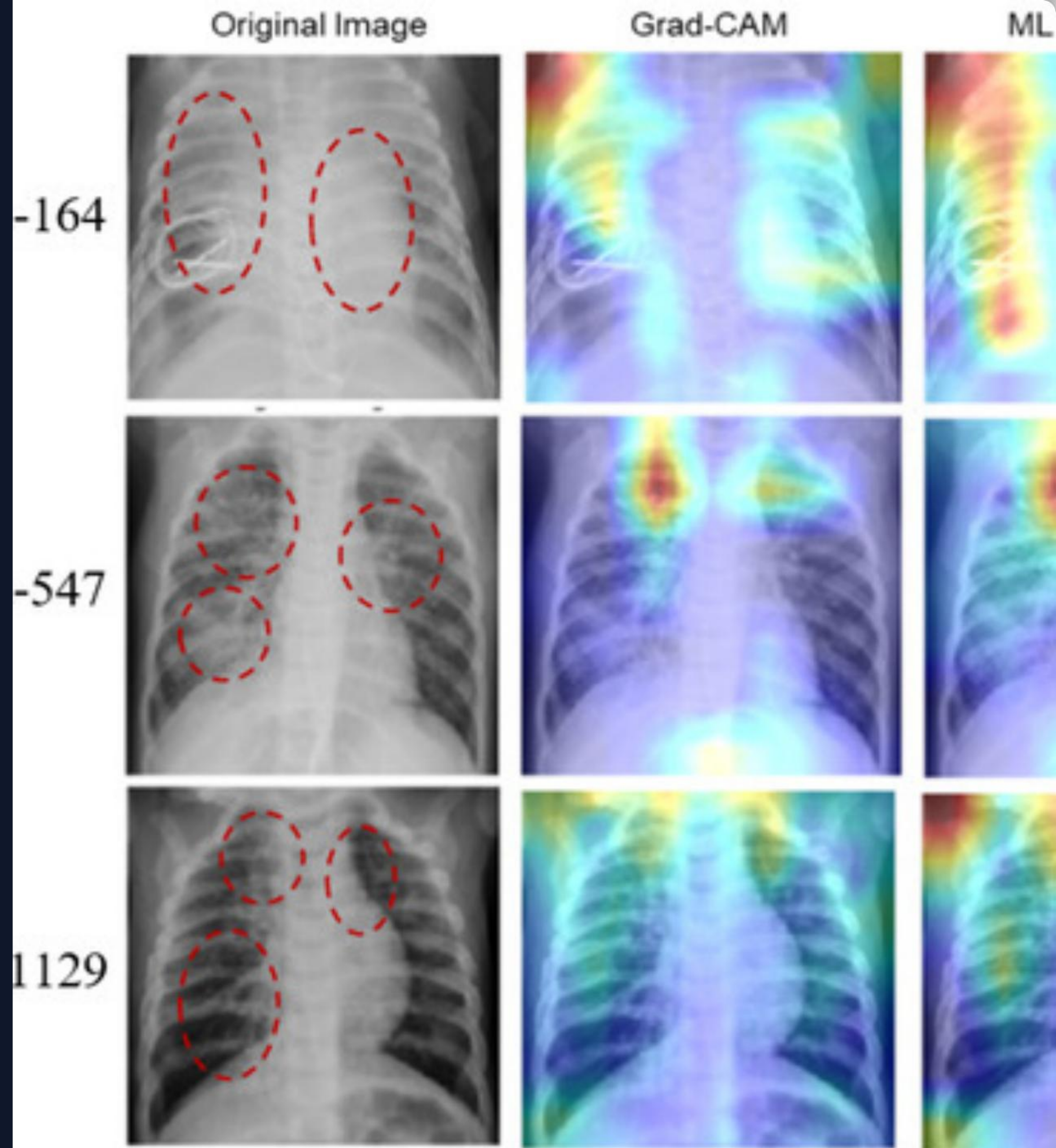
The model is optimized for sensitivity. In clinical triage, missing a pneumonia case is a higher risk than a false alarm. This high-recall profile makes it a dependable clinical companion.

Explainability Audit

Visual Accountability

Grad-CAM allows us to see the "attention map" of the model. This ensures the AI is looking at lung pathology rather than artifacts or text.

Clinical validation confirms that hotspots align with opacified regions identified by experts.



| Current Limitations

-  **Demographic Bias:** Trained primarily on pediatric cohorts; adult anatomy results may vary.
-  **Resolution Loss:** Resizing to 224x224 may lose some micro-structural consolidation details.
-  **Class Overlap:** Binary classification doesn't distinguish between bacterial, viral, or COVID-19 patterns.

Future Roadmaps



Multi-Class

Differentiating bacterial vs viral pneumonia specifically.



High Res

Implementing 512x512 inputs to capture subtle textures.



Multi-Modal

Integrating age, WBC counts, and temperature data.

Conclusion & Questions

A 94.2% sensitivity model ready for clinical diagnostic assistance.

Thank you for your attention!

Image Sources



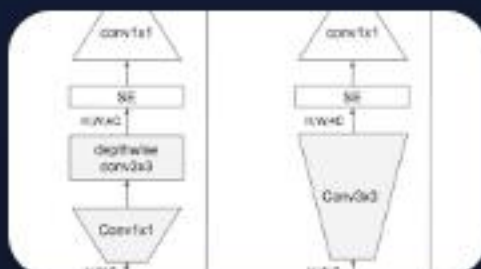
https://png.pngtree.com/thumb_back/fh260/background/20241002/pngtree-x-ray-of-a-person-s-chest-image_16304610.jpg

Source: [pngtree.com](https://png.pngtree.com)



<https://i.ytimg.com/vi/xaz0Uo0SIL4/maxresdefault.jpg>

Source: www.youtube.com



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Source: [medium.com](https://miro.medium.com)



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